Ing. Michal Štěpán

AI & Data Platform Architect | GenAI | Cloud Data Systems

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Introduction

AI and Data Platform Engineer with strong focus on system architecture, GenAI platforms, and scalable data systems.

Experienced in designing end-to-end AI and data solutions across cloud environments, including RAG systems, data pipelines, and enterprise analytics platforms.

Passionate about bridging data engineering, AI capabilities, and business applications to build reliable, production-grade AI systems.

Core skills

Architecture & Systems

- Retrieval-Augmented Generation (RAG)
- Multi-step LLM workflows, tool integration (function calling)
- Agent based systems, orchestration (LangChain, LangGraph)
- Data platform architecture
- Distributed data pipelines

Data & AI Platforms

- Vector databases (Azure Search, ChromaDB, PGVector)
- Document intelligence pipelines
- Model evaluation workflows
- Data quality & observability

Cloud & Infrastructure

- Azure cloud architecture
- Containerized AI services
- CI/CD for data & AI systems

Programming

- Python (FastAPI, Pandas, Pytest)
- SQL

Professional Experience

ABB s.r.o | Senior GenAI Engineer

September 2025 - now

- Designed architecture of a full RAG pipeline (Azure AI Search, vector embeddings, grounding).
- Integration of GenAI into business processes, creation of golden datasets, and model evaluation workflows.
- Development of an internal GenAI application platform using FastAPI, Streamlit, Docker, and Azure Container Apps.
- Deployment and management of containerized applications in Azure Container Apps.
- Administration of Key Vault, Container Registry, Log Analytics.
- Implementation of Document Intelligence and multimodal processing pipelines.

Freelance Software / Data Engineer

September 2022 – September 2025
Projects delivered for DTSE, Landis+Gyr, Vendavo and other clients in energy, telecom and manufacturing.

As a freelance engineer I contributed to multiple long-term projects focused on backend development, data engineering and system automation. My work covered designing modular architectures, creating robust REST services, building data pipelines, and ensuring reliable production operation. I collaborated remotely with international teams and delivered complete solutions from design to deployment.

Key contributions:

- Designed and implemented **end-to-end data workflows** in Python, including orchestration with **Dagster, Airflow**, and data storage in **PostgreSQL, DuckDB, BigQuery**.
- Built and optimized **backend components** for high-volume data ingestion and processing, including message-based processing using **RabbitMQ**.
- Migrated services from **Kubernetes** to **cloud-native deployments** (GCP: Cloud Run, GCS, BigQuery).
- Maintained and improved **CI/CD pipelines** (GitLab CI, GitHub Actions), ensuring high code quality through automated testing (**Pytest: unit, integration, e2e**).
- Delivered high-performance data processing pipelines for recommender systems, including multiprocessing and optimization work.
- Provided L2 engineering support for production systems, diagnosing complex issues and contributing fixes to maintain system reliability.

Siemens s.r.o. | Senior Software Developer

January 2021 – August 2022
Tech Stack: Python, Flask, FastAPI, Docker, PostgreSQL, Grafana, Power BI, RabbitMQ, AWS S3, AWS Lambda

- Developed an Energy Management System to consolidate and analyze data from various sources.
- Implemented ARIMA and SARIMA models for time-series forecasting.
- Mentored junior team members, fostering their technical growth.

ABB s.r.o. | Software Project Engineer

October 2018 – January 2021
Tech Stack: Python, Docker, PostgreSQL, Azure Functions, Azure Blob Storage, Grafana, Power BI

- Development of a water and energy consumption evaluation system.
- Designed and implemented predictive analytics features.

DISTEP a.s. | IT Specialist, Power Engineer, Technical Deputy

May 2010 – October 2018
Tech Stack: MS SQL, VBA

- Streamlined data collection processes using PDA barcode readers.
- Managed water and heat energy distribution operations.
- Enhanced resource efficiency through optimized calculations and monitoring.

Education

Technical University of Ostrava
Master's Degree in Electrical Engineering and Computer Science
Master Thesis: *VBA Utilization for Power Engineering Data Processing and Presentation*

Czech Technical University
Bachelor's Degree in Nuclear Sciences and Physical Engineering
Bachelor Thesis: *Physical Analysis of Fast Reactor's Cores in Up-to-Date Tendencies*

Languages

- Czech: Native Speaker
- English: B2-C1 (FCE Certification)
